Project: Moffat Bay Lodge

Technical Design Document (TDD)

Client: Moffat Bay Lodge

Location: Moffat Bay, Washington

Course Title: CSD-460 Capstone in Software Development

Project: Moffat Bay Lodge Reservation Website

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Date: October 25, 2026

Project Description

Moffat Bay Lodge is a proposed website for lodging reservations, developed as part of our software development capstone project. The goal is to provide visitors with information about the lodge, available rooms, pricing, and nearby attractions. Customers can create an account, log in, make room reservations, and view existing reservations. The website will use Java, JSP, HTML, CSS, and MySQL to manage customer and booking information. Our team will use GitHub for version control and a Kanban board to track development progress.

1. Introduction

The Moffat Bay Lodge project involves developing a website that allows visitors to learn about the lodge, explore nearby attractions, view room options, and make reservations. Customers can register for an account, log in, reserve rooms, and view their reservations.

Our team develops the website using Java, JSP, HTML, CSS, and MySQL. We will use GitHub to manage source code and a Kanban board to organize development tasks. This Technical Design Document will help our team understand the project requirements, identify user needs, and plan the development process.

1.1 Purpose

This project aims to create a user-friendly reservation website for Moffat Bay Lodge. Visitors should be able to browse lodge information without creating an account. However, customers must register and log in before making a reservation.

The website will provide room information, pricing, local attractions, customer registration, login, reservation booking, and reservation lookup features.

The system will store customer and reservation data in a MySQL database. The project does not include online payment processing.

1.2 Terminology

Term	Definition
TDD	Technical Design Document, used to plan the system before development
User Persona	A fictional person representing a type of website user
User Story	A description of what a user wants to accomplish
JSP	JavaServer Pages, used to create dynamic web pages
Java	Programming language used for application logic
MySQL	Database used to store customer and reservation records
HTML	Language used to structure web pages
CSS	Language used to style web pages
Authentication	Process of verifying a user's login information
Session	Server-Side mechanism used to maintain a user's login state
Kanban Board	Tool used to organize track development tasks
Story Hours	Estimated time required to complete development work
ERD	Entity Relationship Diagram showing database relationships

1.3 User Personas

Persona 1: David- New Visitor

Age: 35

Background: David is planning a family vacation and wants to find a comfortable place to stay.

Goals: Explore lodge information, compare room prices, and create an account to make a reservation

Challenges: David wants to understand room options and prices before deciding to book.

Needs: An easy-to-navigate website with clear room information and a simple registration process.

Persona 2: Maria-Registered Customer

Age: 29

Background: Maria enjoys traveling and outdoor activities. She has registered with Moffat Bay Lodge.

Goals: Log in, select a suitable room, choose reservation dates, and confirm her booking.

Challenges: Maria wants to complete the reservation without confusion or unnecessary steps;

Needs: A secure login process, a simple reservation form, and a clear booking summary.

Persona 3: James- Returning Customer

Age: 45

Background: James has previously booked a room at Moffat Bay Lodge and wants to review his reservation information.

Goals: Look up an existing reservation and verify booking details.

Challenges: James may not remember his reservation number and wants another way to find his booking.

Needs: A reservation lookup feature that allows authorized access using reservation **information.**

1.4 User Stories

The following user stories are organized by persona and priority

ID	Persona	User Story	priority
01	David	As a new visitor, I want to register an account so that I can make a reservation.	High-1

02	Maria	As a registered customer, I want to log in securely so that I can access reservation features.	Hign-2
03	Maria	As a registered customer, I want to reserve a room by selecting room type, guests, and dates so that I can plan my stay.	High-3
04	Maria	As a registered customer, I want to review my reservation summary before confirming so that I can verify the details.	High-4
05	Jemes	As a returning customer, I want to look up my reservation so that I can review my booking information	High-5
06	David	As a new visitor, I want to view room options and prices so that I can choose a room within my budget.	High-6
07	David	As a new visitor, I want to learn about lodge amenities and attractions so that I can plan my vacation	High-7
08	James	As a returning customer, I want to view the check-in and check-out dates of my reservation so that I can prepare for my trip.	High-8
09	James	As a returning customer, I want to find the lodge's contact information so that I can ask questions about my stay.	High-9

1.5 Work Estimations (To-Do List)

The three highest-priority user stories are customer registration, customer login, and room reservation.

The following hours are initial team estimates and may change after development begins.

User Story 1: Customer Registration (01)

Goal: Allow new customers to create accounts.

Development task	**Estimated hours**
Create customer registration forms using JSP and HTML	3
Implement email and password validation	3
Create MySQL customer table and database connection	3
Implement secure password hashing and save customer records	4
Test registration and duplicate-email handling	2

Total	15

Registration will collect the customer's first name, last name, email address, telephone number, and password. Passwords will be securely hashed before storage.

User Story 2: Customer Login (02)

Goal: Allow registered customers to log in securely.

Development task	**Estimated hours**
Create login form	2
Implement database lookup and password verification	3
Create authenticated user session and logout process	3
Implement error messages and access restrictions	2
Test valid and invalid attempts	2
Total	12

The login system will verify the customer's email and password against the registered customer information stored in MySQL.

User Story 3: Room Reservation (03)

Goal: Allow authenticated customers to choose a room and reservation dates.

Development task	**Estimated hours**
Create room reservation form	3
Implement room selection and guest input	2
Validate check-in and check-out dates	3
Connect reservation information to the database layer	3
Integrate with reservation summary and confirmation process	3
Test reservation form and booking workflow	3
Total	17

Room prices will be based on the following options:

Room Type	Price per night
Double full beds	$120
Queen	$135
Double queen beds	$150
King	$160

The customer must review the reservation summary and confirm the booking before the system saves the reservation to the MySQL database. The reservation summary feature will be completed as a related user story.

Total Estimated hours for Top Three Stories

User Story	**Estimated hours**
01 Customer Registration	15
02 Customer login	12
03 Room reservation	17
Total	44 hours

These estimates represent development effort rather than calendar time. Some tasks can be completed in parallel with different team members.

Suggested Team Kanban To-Do tasks

The following tasks should be added to the Moffat Bay Tasks GitHub Kanban board:

1. Create customer registration page -3 hours
2. Implement registration validation -3 hours
3. Create customer database table and connection – 3 hours
4. Implement password hashing and customer storage -4 hours
5. Test customer registration -2 hours
6. Create customer login page -2 hours
7. Implement login verification – 3 hours
8. Implement user sessions and logout -3 hours
9. Add login error handling and access restrictions – 2 hours

10. Test customer login – 2 hours

11. Create reservation form -3 hours

12. Implement room and guest selection -2 hours

13. Validate reservation dates -3 hours

14. Connect reservation workflow to MySQL -3 hours

15. Integrate reservation summary and confirmation – 3 hours

16. Test reservation workflow- 3 hours

Kanban Board: Moffat Bay Tasks

The team leader or a designated team member will add these tasks to the To-Do column. Team members will select tasks and update their status as development progresses.

The remaining user stories, including reservation lookup, room information, attractions, and contact information, will be planned and implemented during subsequent development work.

2. Design

Reserved for a future module

2.1 Prototypes

2.2 ERD

3. QA Testing

Reserved for a future module.

3.1 QA Test Plan